

Solution Requirements

Date	30 September 2026
Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

Functional Requirements

ID	Requirement	Description
FR-01	User profile input	Accept user ID, name, age, weight, goal and workout intensity.
FR-02	Personalized workout generation	Generate a structured 7-day plan using Gemini.
FR-03	Nutrition/recovery tip	Generate one concise goal-specific nutrition or recovery tip.
FR-04	Persistence	Store user profile and generated plan in SQLite.
FR-05	Plan update	Accept feedback and generate a revised 7-day plan.
FR-06	Duplicate user handling	Detect an existing User ID and direct the user to the update flow.
FR-07	Admin authentication	Allow an administrator to sign in using environment-configured credentials.
FR-08	Admin dashboard	Display stored users and associated plans to an authenticated admin.
FR-09	Health check	Expose an application health endpoint.

Non-Functional Requirements

Area	Requirement
Usability	Simple responsive HTML interface with readable generated results.
Maintainability	Modular Python files for routes, database, AI generation and authentication.
Security	Secrets stored in environment variables; admin dashboard protected by session state.
Performance	Use lightweight SQLite storage and direct Gemini API calls suitable for a prototype.
Reliability	Return useful error messages for duplicate IDs, missing users and missing plans.